

A counterexample to the conjectured Gagliardo equivalence for one-sided weak fractional Sobolev spaces

Candidate result for arXiv:2004.10833

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Status: candidate counterexample, likely valid. For every $0 < \alpha < 1$ and every $1 < p < \infty$, $p \neq 2$, the conjectured equality on $\mathbb{R}$ is false. The endpoint cases $p = 1, \infty$ are not claimed.

1 Source and exact conjecture

The source is X. Feng and M. Sutton, *A new theory of fractional differential calculus and fractional Sobolev spaces: One-dimensional case*, arXiv:2004.10833v2 [1]. The conjecture appears in Remark 5.19(c), printed page 64 (source line 3960). It is repeated verbatim in the authors' concise follow-up arXiv:2007.10245v1 [2].

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(c) We conjecture that ${}^{\pm}W^{\alpha,p}(\mathbb{R}) \neq \widehat{W}^{\alpha,p}(\mathbb{R})$, but ${}^{\pm}W^{\alpha,p}(\mathbb{R}) = \widetilde{W}^{\alpha,p}(\mathbb{R})$ for $p \neq 2$ and $0 < \alpha < 1$.

(d) It can easily be shown that the equality ${}^{\pm}W^{\alpha,p}(\Omega) = \widetilde{W}^{\alpha,p}(\Omega)$ cannot hold in general. It was proved that when $\alpha p > 1$, ${}^{\pm}\mathcal{D}^{\alpha}C \notin {}^{\pm}W^{\alpha,p}(\Omega)$. However, constant functions always belong to $\widetilde{W}^{\alpha,p}(\Omega)$. In general, $\widetilde{W}^{\alpha,p}(\Omega) \not\subset {}^{\pm}W^{\alpha,p}(\Omega)$. For the same reason, $\widetilde{W}^{\alpha,p}(\Omega) \not\subset W^{\alpha,p}(\Omega)$ when $\alpha p > 1$. This simple example shows that the fractional derivative definition is fundamentally different from the (double) integral term resembling a difference quotient in the seminorm of $\widetilde{W}^{\alpha,p}(\Omega)$. If an equivalence exists on the finite domain, it is our conjecture that for $\alpha p < 1$, the spaces $W^{\alpha,p}(\Omega)$ and $\widetilde{W}^{\alpha,p}(\Omega)$ are the two spaces that should be comparable.

6. WEAK FRACTIONAL DERIVATIVES OF DISTRIBUTIONS

The aim of this section is to introduce some weak fractional derivative notions

Figure 1: The source conjecture on printed page 64 of arXiv:2004.10833v2. This is a direct crop of the archived source PDF.

The relevant clause reads:

“We conjecture that ${}^{\pm}W^{\alpha,p}(\mathbb{R}) \neq \widehat{W}^{\alpha,p}(\mathbb{R})$, but ${}^{\pm}W^{\alpha,p}(\mathbb{R}) = \widetilde{W}^{\alpha,p}(\mathbb{R})$ for $p \neq 2$ and $0 < \alpha < 1$.”

We disprove the asserted equality with $\widetilde{W}^{\alpha,p}(\mathbb{R})$ for every interior exponent $p \neq 2$. We do not address the separate assertion involving $\widehat{W}^{\alpha,p}$.

2 Definitions and proof intuition

For $0 < \alpha < 1$, the source defines ${}^{-}W^{\alpha,p}(\mathbb{R})$ and ${}^{+}W^{\alpha,p}(\mathbb{R})$ by requiring $u \in L^p(\mathbb{R})$ and, respectively, the left or right weak fractional derivative to lie in $L^p(\mathbb{R})$. The Gagliardo space $\widetilde{W}^{\alpha,p}(\mathbb{R})$ consists of the functions $u \in L^p(\mathbb{R})$ for which

$$[u]_{\widetilde{W}^{\alpha,p}(\mathbb{R})}^p = \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{|u(x) - u(y)|^p}{|x - y|^{1+\alpha p}} dx dy < \infty.$$

Proof Intuition

The two norms aggregate dyadic frequencies in different ways. On the line, the left and right fractional derivatives have symbols $(i\xi)^\alpha$ and $(-i\xi)^\alpha$. After removing a harmless Hilbert-transform phase, either graph norm is the Bessel-potential norm $F_{p,2}^\alpha$, whose dyadic pieces are summed in ℓ^2 pointwise. The Gagliardo norm is the Besov norm $B_{p,p}^\alpha$, whose dyadic pieces are summed in ℓ^p . A sequence of widely separated frequency packets with a common spatial envelope turns the two memberships into the elementary alternatives $a \in \ell^2$ and $a \in \ell^p$. These agree only at $p = 2$.

3 The counterexample theorem

Theorem 1. *Let $0 < \alpha < 1$ and $1 < p < \infty$. Then, with equivalent norms,*

$${}^{-}W^{\alpha,p}(\mathbb{R}) = {}^{+}W^{\alpha,p}(\mathbb{R}) = H^{\alpha,p}(\mathbb{R}) = F_{p,2}^\alpha(\mathbb{R}), \quad \widetilde{W}^{\alpha,p}(\mathbb{R}) = B_{p,p}^\alpha(\mathbb{R}).$$

Consequently, if $p \neq 2$, neither one-sided space equals $\widetilde{W}^{\alpha,p}(\mathbb{R})$. More precisely:

- (i) *if $2 < p < \infty$, there exists a real $u \in \widetilde{W}^{\alpha,p}(\mathbb{R}) \setminus ({}^{-}W^{\alpha,p}(\mathbb{R}) \cup {}^{+}W^{\alpha,p}(\mathbb{R}))$;*
- (ii) *if $1 < p < 2$, there exists a real $u \in ({}^{-}W^{\alpha,p}(\mathbb{R}) \cap {}^{+}W^{\alpha,p}(\mathbb{R})) \setminus \widetilde{W}^{\alpha,p}(\mathbb{R})$.*

We split the proof into the operator identification and the packet construction.

Lemma 1 (The source graph spaces are Bessel-potential spaces). *For $0 < \alpha < 1$ and $1 < p < \infty$, each one-sided weak derivative graph norm is equivalent to*

$$\|u\|_{L^p} + \|(-\Delta)^{\alpha/2} u\|_{L^p}.$$

Thus each graph space is $H^{\alpha,p}(\mathbb{R}) = F_{p,2}^\alpha(\mathbb{R})$.

Proof. Use the principal branches

$$(i\xi)^\alpha = |\xi|^\alpha e^{i\theta \operatorname{sgn} \xi}, \quad (-i\xi)^\alpha = |\xi|^\alpha e^{-i\theta \operatorname{sgn} \xi}, \quad \theta = \frac{\pi\alpha}{2}.$$

The source defines the left weak derivative by duality against the right classical derivative, and conversely. Fourier duality on test functions gives

$$\mathcal{F}({}^{-}\mathcal{D}^\alpha u)(\xi) = (i\xi)^\alpha \widehat{u}(\xi), \quad \mathcal{F}({}^{+}\mathcal{D}^\alpha u)(\xi) = (-i\xi)^\alpha \widehat{u}(\xi)$$

in tempered distributions. This identity is particularly unambiguous for the $L^1 \cap L^p$ witnesses constructed below; conversely an L^p inverse Fourier multiplier satisfying the identity is the source weak derivative by the defining test relation.

Let H be the Hilbert transform, with multiplier $-i \operatorname{sgn} \xi$. The phase multipliers above are

$$e^{\pm i \theta \operatorname{sgn} \xi} = \cos \theta \mp \sin \theta \widehat{H}(\xi).$$

They and their inverses are bounded on L^p by the M. Riesz theorem [3]. It follows that

$$\| \mathcal{D}^\alpha u \|_p \asymp \| (-\Delta)^{\alpha/2} u \|_p \asymp \| {}^+ \mathcal{D}^\alpha u \|_p.$$

The inhomogeneous graph norm on the right is equivalent, by the usual smooth multiplier comparison, to $\| (1 - \Delta)^{\alpha/2} u \|_p$, which defines $H^{\alpha,p}$. The classical Littlewood–Paley characterization gives $H^{\alpha,p} = F_{p,2}^\alpha$ [4]. $\square$

Lemma 2 (A sparse packet separates the sequence exponents). *Fix $0 \neq \eta \in \mathcal{S}(\mathbb{R})$ whose Fourier transform is supported in a sufficiently small interval about the origin. Put $n_k = 4k$. For every polynomially bounded scalar sequence $a = (a_k)$, the series*

$$u_a(x) = \sum_{k=1}^{\infty} a_k, 2^{-\alpha n_k} e^{i 2^{n_k} x} \eta(x) \quad (1)$$

converges in $L^1(\mathbb{R}) \cap L^p(\mathbb{R})$ and satisfies

$$u_a \in F_{p,2}^\alpha(\mathbb{R}) \iff a \in \ell^2, \quad u_a \in B_{p,p}^\alpha(\mathbb{R}) \iff a \in \ell^p. \quad (2)$$

Proof. The factor $2^{-\alpha n_k}$ makes (1) absolutely summable in both L^1 and L^p . The Fourier supports of its terms lie in pairwise separated dyadic annuli. Choose smooth projections Q_k , uniformly obtained by dilation, which equal one on those supports and vanish on the other packet supports. Admissible Littlewood–Paley norms are independent of the chosen smooth dyadic resolution [4]. Hence the sparse square-function and Besov norms are, with constants independent of a ,

$$\begin{aligned} \|u_a\|_{F_{p,2}^\alpha} &\asymp \|u_a\|_p + \left\| \left(\sum_k |2^{\alpha n_k} Q_k u_a|^2 \right)^{1/2} \right\|_p \\ &= \|u_a\|_p + \|\eta\|_p \left(\sum_k |a_k|^2 \right)^{1/2}, \\ \|u_a\|_{B_{p,p}^\alpha} &\asymp \|u_a\|_p + \left(\sum_k 2^{\alpha p n_k} \|Q_k u_a\|_p^p \right)^{1/p} \\ &= \|u_a\|_p + \|\eta\|_p \left(\sum_k |a_k|^p \right)^{1/p}. \end{aligned}$$

This proves (2). Equivalently, one may insert the Q_k into any fixed dyadic resolution; finite overlap and the vector-valued multiplier theorem give the same two estimates. $\square$

Proof of Theorem 1. The first identity is Lemma 1. The classical difference characterization of Besov spaces gives $\widetilde{W}^{\alpha,p}(\mathbb{R}) = B_{p,p}^\alpha(\mathbb{R})$ for $0 < \alpha < 1$ and $1 < p < \infty$ [4].

Set

$$\beta = \frac{1}{2} \left(\frac{1}{2} + \frac{1}{p} \right), \quad a_k = k^{-\beta}.$$

If $p > 2$, then $1/p < \beta < 1/2$, so $a \in \ell^p \setminus \ell^2$. Lemma 2 gives $u_a \in B_{p,p}^\alpha \setminus F_{p,2}^\alpha$. If $p < 2$, then $1/2 < \beta < 1/p$, so $a \in \ell^2 \setminus \ell^p$, and the reverse membership holds.

The displayed witness is complex-valued only for notational convenience. Both spaces are stable under real and imaginary parts. If both components belonged to the excluded space, then their complex sum would also belong to it. Thus at least one of $\operatorname{Re} u_a$ and $\operatorname{Im} u_a$ is a real-valued witness with the stated membership. $\square$

4 Verification, limitations, and novelty

The proof uses four classical tools: the L^p bound for the Hilbert transform; the identity $H^{\alpha,p} = F_{p,2}^\alpha$; the difference description $\widetilde{W}^{\alpha,p} = B_{p,p}^\alpha$; and invariance under the choice of smooth dyadic resolution. These are all standard for interior p [3, 4]. No finite computation can verify the separation: every finite packet sum lies in both spaces, and the obstruction is exactly the infinite ℓ^2 versus ℓ^p summability.

The endpoint cases $p = 1$ and $p = \infty$ remain outside the theorem because the Hilbert transform is not bounded on those L^p spaces. We also make no claim about the source's nonstandard Fourier- L^p space $\widehat{W}^{\alpha,p}$, nor about finite intervals in Remark 5.19(d).

A bounded search (26 August 2026) covered the run's registry, solution, attempt, and proof-gap indexes; the exact wording of Remark 5.19(c); the repeated arXiv:2007.10245 wording; exact-title, author, and core-term arXiv queries; and the later visible Feng–Sutton arXiv work. No later paper was found that explicitly claims to resolve this exact conjecture. Novelty confidence is therefore moderate, not high: the general distinction between $F_{p,2}^\alpha$ and $B_{p,p}^\alpha$ is classical. The candidate contribution is the exact identification of the source graph spaces and the explicit application that disproves its conjecture for all interior $p \neq 2$.

Human review recommendation. Send to a harmonic analyst or function-space expert. The most useful checks are the Fourier-duality convention in Lemma 1 and the sparse projection norm equivalence in Lemma 2.

References

- [1] X. Feng and M. Sutton, *A new theory of fractional differential calculus and fractional Sobolev spaces: One-dimensional case*, arXiv:2004.10833v2 (2020).
- [2] X. Feng and M. Sutton, *On New Families of Fractional Sobolev Spaces*, arXiv:2007.10245v1 (2020).
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- [4] H. Triebel, *Theory of Function Spaces*, Birkhäuser, 1983.